FINAL PROJECT · AI-BASED SOLUTION

EXPLAINABLE AI FOR VERIFYING PLASTIC PACKAGING RECYCLING CLAIMS

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ENVIRONMENTAL CLAIMS ≠ VERIFIED EVIDENCE

Manual ESG reviews are time-consuming; plastic recycling disclosures are often fragmented and vary in scope.

CHALLENGE

“100% recyclable”

A claim may lack definitions, time frames, scope boundaries, or quantitative support.

One statement alone cannot establish greenwashing.

1 WHAT

What exactly does the company claim about plastic packaging?

2 WHERE

Which passages, tables, and pages support that claim?

3 CHECK

Do the scope, year, units, and reported rates agree?

PROPOSED SAMPLE: 20–30 PUBLIC REPORTS

This is a proposed evaluation corpus, not a dataset already collected or used to train a model.

20–30

PDF REPORTS

Sustainability / ESG / Environmental disclosures

1 **SOURCES**

Official company websites; sustainability, ESG, and environmental report PDFs.

2 **COMPANIES**

FMCG firms, potentially Vinamilk, Masan Consumer, and Unilever, subject to data availability.

3 **ELIGIBILITY**

Relevant disclosures on plastics, recyclability, collection, recycling, rPET, or targets.

Selection criteria: publicly available • readable • traceable company/year/page • relevant packaging content

WHAT DOES THE CRAWLER COLLECT?

The crawler downloads documents and metadata; LLMs interpret claims and evidence in subsequent stages.

01

SEED

Company sustainability disclosure pages

02

DISCOVER

Relevant public report PDF links

03

DOWNLOAD

PDF files; detect duplicates and corrupted files

04

CATALOG

Company, year, URL, file name, and page count

OUTPUT OF CRAWLER

`/raw_pdfs/*.pdf`

`manifest.csv → company | year | report_type | source_url | pdf_path`

CONVERT PDFS INTO SEARCHABLE EVIDENCE

Preserve original page references and table context so that every LLM assessment can be audited.

SOURCE PDF

Text + tables + scanned pages

Original document

›

PARSER / OCR

Extract text, tables, page numbers

OCR for scans

›

CHUNK + INDEX

Split by section/page; attach metadata

Build retrieval index

EXAMPLE OUTPUT CHUNK

```
company=FMCG_A | year=2025 | page=42 | section=Packaging
```

Keep table values, footnotes, and measurement units attached to their source context.

WHAT DOES THE LLM READ AND EXTRACT?

Focus on plastic packaging recyclability, actual recycling performance, recycled content, and future commitments.

1. **RECYCLABILITY**
   Claims that packaging can be recycled

2. **PERFORMANCE**
   Amount of plastic collected or recycled

3. **RECYCLED CONTENT**
   Share of recycled plastic in packaging

4. **TARGET / COMMITMENT**
   Future objectives and deadlines

STRUCTURED CLAIM


```
{
  "claim_text": "100% recyclable by 2025",
  "claim_type": "target",
  "company": "FMCG_A",
  "report_year": 2024,
  "target_year": 2025,
  "material": "plastic packaging",
  "scope": "not specified",
  "source_page": 18
}
```


ILLUSTRATIVE EXAMPLE — NOT REAL DATA

FROM CLAIMS TO EVIDENCE-BASED VERIFICATION

The LLM must evaluate retrieved evidence and return traceable references to source pages.

CLAIM

Structured claim record

›

RETRIEVE

Top-k relevant passages/tables

›

COMPARE

LLM evaluates defined criteria

›

EXPLAIN

Label + reason + citation

FOUR VERIFICATION CRITERIA

- **SAME YEAR?** — Reporting year versus target year
- **SAME SCOPE?** — Geography, products, materials
- **SUFFICIENT?** — Figures, methods, definitions
- **CONTRADICTIONS?** — Conflicting figures or contexts

Missing evidence ≠ false claim

PYTHON RECALCULATES METRICS — NOT THE LLM

Only compare values with compatible company, year, material, geography, scope, and denominator.

10,000 t

PACKAGING PLACED

Packaging placed on the market

4,000 t

COLLECTED

Packaging collected

2,500 t

RECYCLED

Packaging actually recycled

40% — **Collection rate**

4,000 / 10,000

25% — **Recycled / placed**

2,500 / 10,000

62.5% — **Recycled / collected**

2,500 / 4,000

CAUTION: Both 25% and 62.5% are mathematically correct, but use DIFFERENT DENOMINATORS.

Illustrative figures adapted from the proposal; they do not describe any specific company.

FOUR ASSESSMENT LABELS FOR HUMAN REVIEW

These are preliminary, evidence-grounded assessments — not definitive findings of greenwashing.

SUPPORTED

Evidence adequately supports the claim

Claim and evidence align in meaning, time, and scope.

INSUFFICIENT EVIDENCE

Not enough evidence to verify

No adequate support found; this does not mean the claim is false.

SCOPE MISMATCH

Evidence applies to a different scope

For example, a global claim supported only by country-level data.

POTENTIAL INCONSISTENCY

Conflicting information detected

Passages or tables disagree on figures, definitions, or context.

HUMAN IN THE LOOP: Review ambiguous and conflicting cases

WHAT DOES THE SYSTEM PRODUCE FOR EACH CLAIM?

The Streamlit prototype lets reviewers trace every assessment back to the original report page.

CLAIM REVIEW / FMCG_A / REPORT 2024 DEMO UI

Claim #014

“100% of plastic packaging will be recyclable by 2025.”

INSUFFICIENT EVIDENCE · MOCKUP

RETRIEVED EVIDENCE

The report states a packaging design target, but provides no table confirming achievement or the scope covered.

Source: Report 2024 · p.18 · “Packaging strategy”

01 **Claim dataset**

claim, company, year, scope, page

02 **Verification result**

label, rationale, evidence_refs

03 **Numeric flags**

formula, denominator, discrepancy

04 **Export**

CSV / JSON + dashboard

Illustrative UI and citations; not actual experiment results.

How Will We Evaluate the 20–30 Reports?

Create human-annotated reference labels and evaluate each pipeline component separately.

1

CLAIM EXTRACTION

Precision · Recall · F1

Does the system identify relevant plastic recycling claims?

2

EVIDENCE RETRIEVAL

Recall@k · Citation accuracy

Are the evidence passages and cited pages correct?

3

VERIFICATION LABEL

Macro-F1 · Confusion matrix

Do the four labels agree with human assessments?

4

NUMERICAL CHECK

Rule pass / fail review

Are formulas, units, and denominators accurate?

DELIVERABLES

Human-labeled claim dataset • Python pipeline • Streamlit prototype • Evaluation report

Workflow: Collect reports → annotate claims → run pipeline → evaluate against labels → fix errors.

THANK YOU

Questions & Discussion

Explainable AI · Evidence-based ESG review